



Smart Lab

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- 1- PIR Motion Detection Subsystem*
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4.1 PIR Motion Detection Subsystem

4.1.1 Introduction and Objective

The Passive Infrared (PIR) Motion Detection Circuit represents a foundational environmental telemetry unit within the Smart Lab ecosystem. The primary objective of this subsystem is to provide real-time, non-intrusive laboratory occupancy tracking to achieve intelligent automation, enhanced security enforcement, and optimal energy conservation. By continuously acquiring spatial occupancy data, the centralized main controller can dynamically govern high-voltage appliances—such as room lighting and ventilation fans—ensuring they remain operational strictly during human presence.

4.1.2 Component Specifications

To realize this subsystem, a selection of specific hardware components was deployed. Table 4.1 outlines the primary hardware layer utilized in the motion detection module:

Component	Quantity	Technical Function / Role
ESP8266 NodeMCU	1	Serves as the localized edge-computing Node; processes raw digital signals and transmits telemetry data wirelessly via HTTP.
HC-SR501 PIR Sensor	1	Detects human motion by sensing dynamic variations in the ambient infrared radiation spectrum.
Jumper Wires	Several	Establishes secure electrical point-to-point connections between the sensor and the microcontroller pins.
USB Data Cable	1	Delivers a stable 5V DC power supply to the microcontroller board and

		facilitates firmware flashing.
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Table 4.1: Hardware Component Stack for the PIR Subsystem

4.1.3 Circuit Design and Interconnections

The hardware integration relies on a localized distributed architecture where the sensor directly interfaces with the digital peripheral pins of the ESP8266 NodeMCU. The operational power is derived from the microcontroller's regulated power rails.

The exact pin configuration mapping implemented for the physical circuit is detailed comprehensively in [Table 4.2](#)

PIR Sensor Pin	NodeMCU Pin Identifier	Electrical Purpose / Signal Direction
VCC	3.3V / Vin	Delivers the positive operating voltage to power the sensor's internal circuitry.
GND	GND	Establishes a common electrical ground reference across the subsystem.
OUT	D1 (GPIO5)	Transmits a digital signal (HIGH upon motion detection; LOW during idle states) to the controller.

Table 4.2: Pin Mapping for the HC-SR501 and ESP8266 NodeMCU Interconnection

The schematic design and the corresponding layout for the motion detection circuit are illustrated in the [figures](#) below:

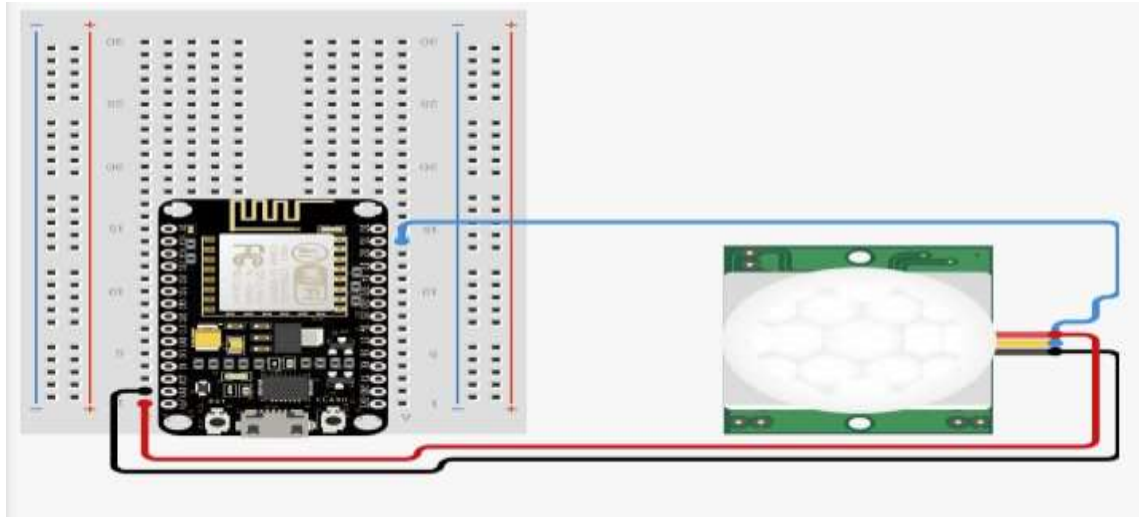


Figure 4.1: Schematic and Breadboard Connection Diagram of the PIR Sensor Interfaced with ESP8266 NodeMCU

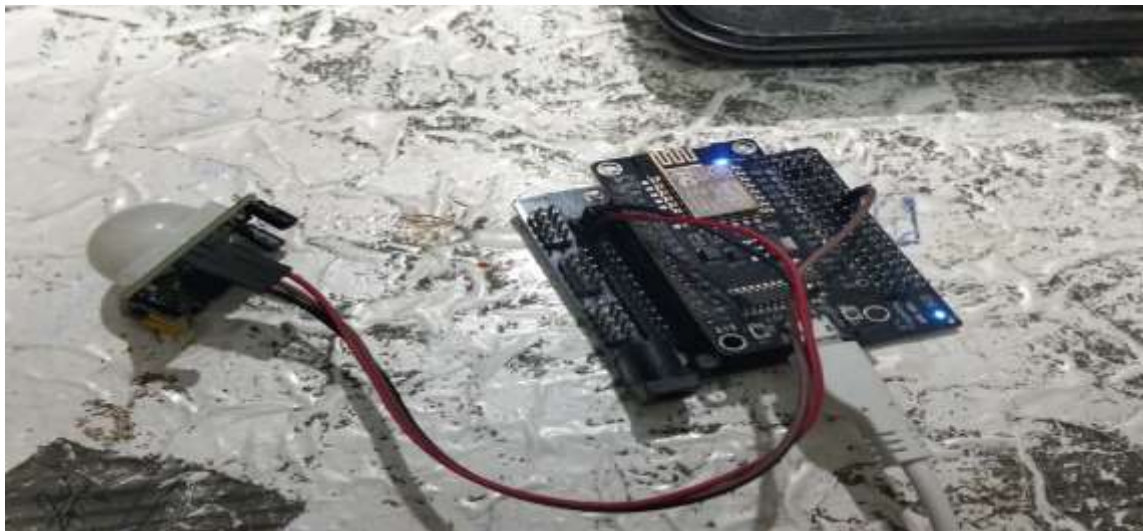


Figure 4.2: Physical Hardware Implementation and Wiring Setup of the PIR Sensing Node during Operation

4.1.4 Working Principle and System Integration

The functional execution of the occupancy sensing unit operates according to a structured cyclic process:

Infrared Pattern Monitoring: The HC-SR501 sensor continuously scans the ambient laboratory space, establishing a baseline thermal profile of localized infrared radiation.

Signal Generation: When a human body moves through the detection cone, it alters the infrared pattern, prompting the sensor to output a digital HIGH signal

(+3.3V).

Edge Processing & Wireless Telemetry: The NodeMCU detects this state change via its digital input pin and immediately structures an asynchronous communication package. This package is transmitted over the local Wi-Fi Access Point using lightweight HTTP requests directed to the Main Web Server.

Interface Updating: The central server processes the telemetry, instantly updates the web-based HTML Dashboard status to display an active alert banner, and evaluates automation rule-base logic to actuate relevant relays.

4.1.5 Experimental Results and Performance Analysis

Empirical testing under varying laboratory configurations and environmental noise levels validated the high reliability of the subsystem. The experimental outcomes demonstrated:

Zero communication latency degradation during continuous inter-board data synchronization over the localized Wi-Fi network.

High sensitivity in human motion detection within the laboratory environment, combined with stable execution of the associated automation rules.

By calibrating the onboard potentiometers on the HC-SR501 module, the physical hardware achieved an optimized balance between rapid response times and the complete elimination of false-positive triggers caused by static ambient heat sources.

4.1.6 Engineering Evaluation: Advantages and Limitations

The selection of the HC-SR501 module for the Smart Lab system is justified by an evaluation of its engineering parameters:

Key Advantages: Exceptionally low power consumption during idle periods, highly cost-effective deployment, direct binary digital output simplifying edge-code execution, and adjustable sensitivity and output delay settings.

Inherent Constraints: The sensor is strictly dependent on relative physical movement; consequently, completely stationary occupants cannot be continuous

detected. Additionally, the standalone circuit is incapable of resolving the exact count of individuals present within the monitored sector.

4.2 DHT11 Temperature and Humidity Monitoring Subsystem

4.2.1 Introduction and Operational Objective

The Temperature and Humidity Monitoring Subsystem serves as the core environmental telemetry unit within the Smart Lab ecosystem. Maintaining strict control over ambient atmospheric profiles is vital in academic and research laboratories, as sensitive electronic instruments, computing nodes, and experimental components demand regulated thermal ranges for safe, optimal, and continuous operation.

The primary objective of this subsystem is to provide non-stop ambient monitoring, feeding real-time data to the centralized automation architecture. This enables micro-decisions such as safety-driven logic overrides, proactive HVAC or fan actuation, and immediate warning flags upon thermal threshold violations.

4.2.2 Subsystem Component Configuration

The physical execution of this telemetry unit is sustained by the hardware components outlined in [Table 4.3](#):

Component	Quantity	Technical Role and Function
ESP8266 NodeMCU	1	Operates as the edge sensory node; handles single-wire data processing and maintains Wi-Fi telemetry sync.
DHT11 Sensor Module	1	Captures ambient relative humidity and temperature using internal capacitive and thermistor mechanisms.
Jumper Wires	Several	Establishes physical point-to-point electrical lines for data and power transmission.

USB Data Cable	1	Delivers a stable +5V DC input to the microcontroller and assists in diagnostic programming.
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Table 4.3: Hardware Configuration Stack for the DHT11 Subsystem

4.2.3 Circuit Design and Hardware Interconnections

The hardware design involves direct interfacing between the DHT11 module and the digital input layout of the ESP8266 NodeMCU. Because the DHT11 incorporates internal signal conditioning and on-board calibration, it provides a pre-processed digital output. This setup removes the need for complex analog-to-digital conversions on the main controller and guarantees data integrity over the data lines.

The exact pin map routing executed for this physical circuit is documented in [Table 4.4](#):

DHT11 Sensor Pin	NodeMCU Pin Designation	Electrical Subsystem Purpose / Signal Path
VCC	3.3V	Sources the regulated operational voltage to power the sensor module
GND	GND	Sets the common ground reference across the local micro-circuitry.
DATA	D4 (GPIO2)	Transmits the synchronized digital packets via a single-wire protocol.

Table 4.4: Pin Configurations for the DHT11 and ESP8266 NodeMCU Interconnection

The detailed schematic wiring layout and the running physical prototype are presented in the following figures:

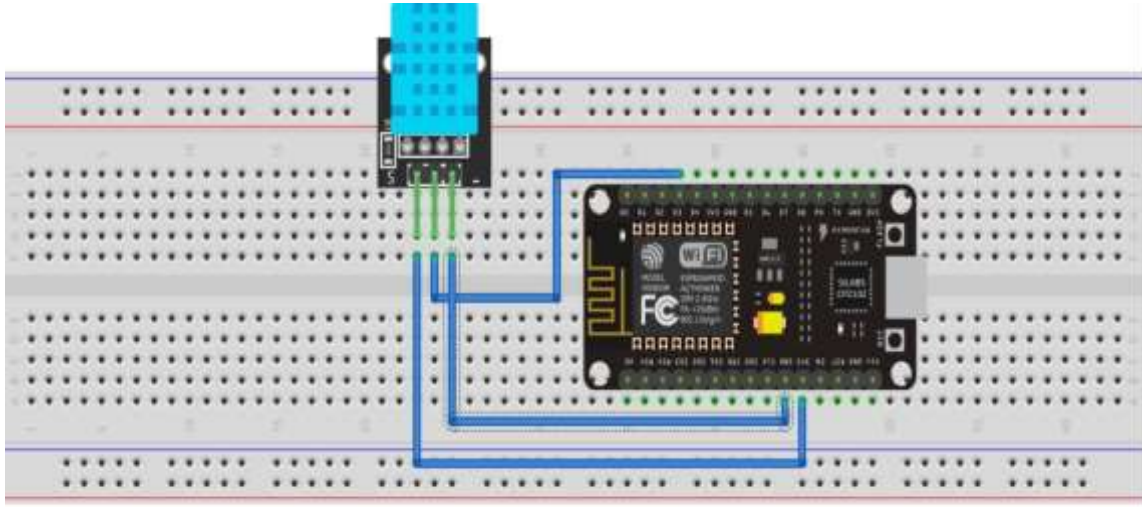


Figure 4.3: Schematic and Breadboard Layout of the DHT11 Sensor Interfaced with ESP8266 NodeMCU



Figure 4.4: Real-world Hardware Assembly and Experimental Running State of the Environmental Telemetry Node

4.2.4 Working Principle and System Integration

The operational workflow of the environmental monitoring subsystem proceeds through a highly structured cycle:

Sensor Initialization: The NodeMCU sends an initial start signal to activate the DHT11 sensing components.

Data Acquisition and Calibration: The module uses its built-in capacitive moisture sensor and NTC thermistor to capture ambient parameters,

automatically matching them with factory-set calibration data.

Single-Wire Transmission: The calibrated data is structured into a 40-bit digital packet and streamed through the single-wire DATA line directly to the NodeMCU.

Wireless Synchronization: The local node reads the incoming data packet, confirms its checksum validity, and forwards the data over the local Wi-Fi Access Point using lightweight HTTP requests.

Dashboard Rendering & Rule Execution: The Main Server parses the payload, updates the web-based HTML Dashboard instantly (displaying current metrics such as 34.1°C temperature and 26% relative humidity), and checks the threshold conditions. For instance, if the temperature matches or surpasses the critical 30°C threshold, the system initiates safety overrides to actuate the cooling fan relay automatically.

4.2.5 Experimental Analysis and Practical Outcomes

Prolonged empirical testing under steady-state and dynamic laboratory conditions verified the resilience of the environmental subsystem. The testing phase highlighted several positive outcomes:

Constant data synchronization with a stable update frequency and zero communication dropouts.

Immediate UI rendering on the HTML Dashboard layout, reflecting environmental fluctuations accurately.

Reliable trigger responses from the automation safety rules, confirming the circuit's suitability for smart educational laboratory ecosystems.

4.2.6 Engineering Evaluation: Justification and Limitations

Engineering Advantages: The DHT11 was chosen due to its robust digital output, simple three-pin implementation, low power consumption, budget-friendly cost, and excellent support from native Arduino IDE libraries. It measures both critical atmospheric factors over a single microcontroller

terminal.

Inherent Constraints: The module features moderate resolution and accuracy, which makes it less ideal for industrial-grade cleanrooms or highly sensitive chemical environments. It also has a slow sampling rate (1 Hz), meaning it requires up to one second between successive data requests. However, for smart building automation and educational applications, its parameters are perfectly optimized.

4.3 MQ-2 Hazardous Gas and Smoke Detection Subsystem

4.3.1 Introduction and Operational Objective

The Hazardous Gas and Smoke Detection Subsystem serves as the primary preventive life-safety unit within the Smart Lab architecture. In educational and research engineering laboratories, the sudden leakage of volatile combustible gases or the accumulation of toxic smoke poses a direct threat to high-value hardware infrastructure and human life.

The primary operational objective of this subsystem is to provide non-stop atmospheric air quality surveillance by acquiring gas concentration levels in real time. This early warning layer communicates directly with the centralized controller to trigger automatic emergency protocols, contain fire hazards, and activate emergency ventilation lines long before dangerous localized thresholds are surpassed.

4.3.2 Hardware Composition Stack

The complete sensory deployment for the industrial safety and air monitoring node utilizes the hardware components listed in Table 4.5:

Component	Quantity	Technical Role and Operational Description
ESP8266 NodeMCU	1	Acts as the local edge-computing unit; captures analog sensory data, executes

		local threshold comparison, and manages Wi-Fi payload delivery.
MQ-2 Gas Sensor Module	1	A tin-dioxide (SnO ₂) semiconductor sensor that tracks the presence and concentration of combustible gases, liquefied petroleum gas (LPG), propane, methane, hydrogen, and industrial smoke
Jumper Wires	Several	Establishes critical electrical connections between the sensor module pins and the microcontroller board.
USB Data Cable	1	Facilitates continuous +5V DC electrical power delivery to the system and enables serial debugging data transfer.

Table 4.5: Hardware Component Specifications for the MQ-2 Subsystem

4.3.3 Circuit Design and Hardware Layout

The electrical configuration integrates the MQ-2 module directly with the analog peripheral layer of the ESP8266 NodeMCU. The MQ-2 sensor operates on electrochemical principles; its internal SnO₂ semiconductor coating changes its electrical resistance upon contact with flammable gases or smoke. This resistance shift produces an analog voltage output directly proportional to the localized gas concentration.

This output is routed to the dedicated analog input terminal (A0) of the NodeMCU. This single-channel analog layout enables the microcontroller to continuously sample fluctuating voltages and convert them into raw digital readings for processing.

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The detailed physical pin map assignment used to implement this hardware safety circuit is outlined in [Table 4.6](#):

MQ-2 Sensor Pin	NodeMCU Pin Identifier	Electrical Connection Purpose / Signal Direction
VCC	3.3V / Vin	Delivers the positive operating voltage to power the sensor's internal heating element and chemistry layer.
GND	GND	Completes the circuit loop by establishing a common ground reference across the nodes.
A0	A0	Routes the raw analog voltage signal representing real-time gas/smoke concentration levels.

Table 4.6: Pin Configurations for the MQ-2 and ESP8266 NodeMCU Interface

The visual schematic routing and the physical operational assembly for the gas monitoring node are shown in the [figures](#) below:

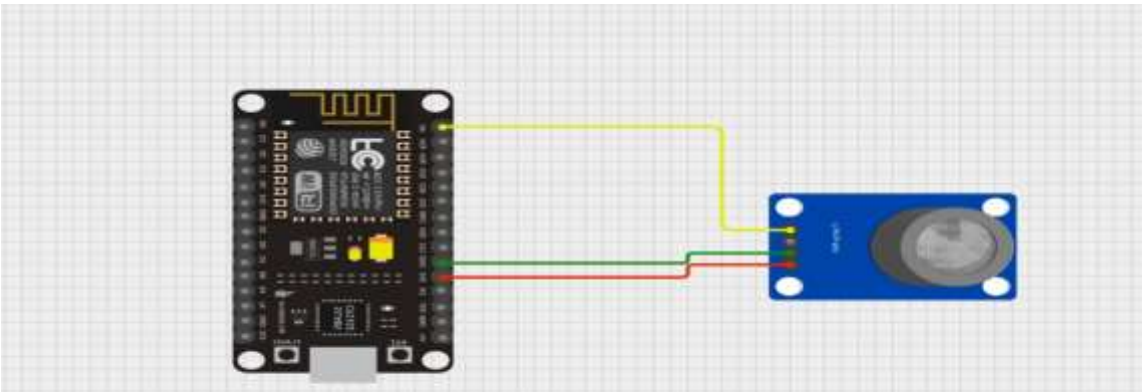


Figure 4.5: Electrical Schematic and Interconnection Diagram of the MQ-2 Gas Sensor Module with the ESP8266 NodeMCU



Figure 4.6: Physical Hardware Assembly and Real-World Testing Layout of the Gas and Smoke Detection Sensory Node

4.3.4 Working Principle and System Integration

The execution of the automated air-safety subsystem follows a deterministic sequential cycle during operation:

Sensor Stabilization & Monitoring: Upon receiving power, the MQ-2 sensor runs its internal heating element to achieve stable operating conditions, thereafter monitoring the laboratory air profile continuously.

Electrochemical Conduction: When smoke or combustible gas encounters the sensor surface, a reduction reaction increases the material's conductivity, causing the analog output pin (AO) to emit an elevated voltage.

Analog-to-Digital Processing: The NodeMCU reads this changing voltage via its internal 10-bit Analog-to-Digital Converter (ADC) pin A0, translating the voltage into a digital scale from 0 to 1023.

Wireless Data Telemetry: The local edge node encapsulates this reading and sends an asynchronous HTTP request over the local Wi-Fi Access Point to update the Main Server.

Dashboard Rendering & Critical Relays Override: The central server instantly parses the numeric concentration level (e.g., displaying a current safe baseline value of 447 with a green "Safe Level " status indicator) on the web-based HTML Dashboard. If a hazardous gas leakage alert occurs, the safety-first override logic takes immediate priority, commanding the 6-channel relay board to instantaneously cut off power to all electrical relays to prevent electrical fire hazards, while concurrently forcing the emergency ventilation fan to active manual override status.

4.3.5 Experimental Results and Performance Analysis

Empirical testing was carried out under controlled conditions using localized smoke and gas sources to evaluate system performance. The experimental outcomes validated:

High proportional sensitivity, as the raw analog digital values shifted smoothly and dynamically in response to varying smoke densities.

Zero communication latency dropouts during continuous data transmission to the main controller dashboard over the local Wi-Fi link.

Immediate and successful execution of critical safety logic rules—specifically the automated relay shut-off protocol—confirming the circuit's effectiveness as an early warning system for smart laboratory spaces.

4.3.6 Engineering Evaluation: Advantages and Limitations

Engineering Advantages: The MQ-2 sensor provides excellent multi-gas sensitivity at a very low implementation cost, offers flexible threshold calibration via its raw analog output, integrates easily with the ESP8266 platform, and delivers reliable performance for smart building safety applications.

Inherent Constraints: The module requires a brief warm-up period to stabilize its heating element before producing accurate data. As a broad-spectrum sensor, it cannot isolate or identify the exact chemical type of gas detected. Additionally, its analog values are sensitive to ambient temperature and humidity shifts, meaning it is intended for general safety overrides rather than precise industrial-grade chemical gas analysis.

4.4 Multi-Channel Relay Actuation and Control Subsystem

4.4.1 Introduction and Operational Objective

While data-acquisition subsystems focus on ambient sensing, the Multi-Channel Relay Control Subsystem operates as the primary physical execution and actuation layer within the Smart Lab ecosystem. Raw sensory data must be translated into automated spatial corrections; the relay module bridges this gap by acting as an electrically operated isolation switch that handles high-voltage power routing based on low-voltage logic commands.

The primary operational objective of this subsystem is to provide autonomous and remote load switching for laboratory appliances. By interfacing the microcontroller with electromagnetic relays, the system can automatically actuate AC/DC equipment—such as auxiliary ventilation fans, primary lighting

grids, and high-intensity warning alarms—in response to real-time telemetry or direct user commands via the web interface, all while maintaining safe galvanic isolation.

4.4.2 Hardware Interconnection Components

The physical deployment of the actuation layer is implemented using the precise hardware component arrangement outlined in Table 4.7:

Component	Quantity	Technical Role and Operational Description
ESP8266 NodeMCU	1	Serves as the central decision-making edge node; processes conditional logic overrides and generates discrete digital output signals
4-Channel Relay Module	1	Contains 4 independent electromechanical relays equipped with optocouplers to safely switch high-current electrical loads
Jumper Wires	Several	Establish reliable electrical pathways between the control terminals of the NodeMCU and the actuation module.
Peripheral Infrastructure Load	As Req	Represents appliances under automated control (e.g., lighting grids, exhaust fans, and safety indicators).
USB Power Interface	1	Supplies stable operational voltage to the microcontroller terminal while serving as a serial diagnostic interface.

Table 4.7: Hardware Actuation Stack for the Relay Subsystem

4.4.3 Circuit Topology and Hardware Interconnections

The hardware layout involves routing physical digital output pins from the ESP8266 NodeMCU to the logic inputs (IN1–IN4) of the 4-channel relay board. The relay module features an integrated optoisolation stage (optocouplers), which ensures that the low-power microcontroller is completely protected from inductive voltage spikes (flyback voltage) generated when the internal electromagnetic relay coils collapse during switching.

The physical load appliances are wired directly through the power terminal blocks using the Normally Open (NO) and Common (COM) configuration. This configuration ensures that the high-voltage circuit remains safely broken and de-

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energized until the NodeMCU explicitly drives a logic command to close the circuit.

The precise physical pin mapping mapped across the prototype wiring is documented in [Table 4.8](#):

Relay Module Pin Terminal	NodeMCU Pin Identifier	Electrical Subsystem Purpose / Routing Protocol
VCC	VIN (5V Line)	Sources +5V DC power to energize the onboard electromagnetic switching coils.
GND	GND	Establishes a common reference ground across the control board infrastructure
IN1	D1 (GPIO5)	Drives the digital actuation path to activate Relay Channel 1 (e.g., Lighting Control).
IN2	D2 (GPIO4)	Drives the digital actuation path to activate Relay Channel 2 (e.g., Ventilation Fan).
IN3	D5 (GPIO14)	Drives the digital actuation path to activate Relay Channel 3 (e.g., Warning Siren).
IN4	D6 (GPIO12)	Drives the digital actuation path to activate Relay Channel 4 (e.g., Auxiliary Systems).

Table 4.8: Peripheral Interconnection Matrix for Actuation Control

The physical assembly and running configuration of this multi-channel actuation circuit are presented below:

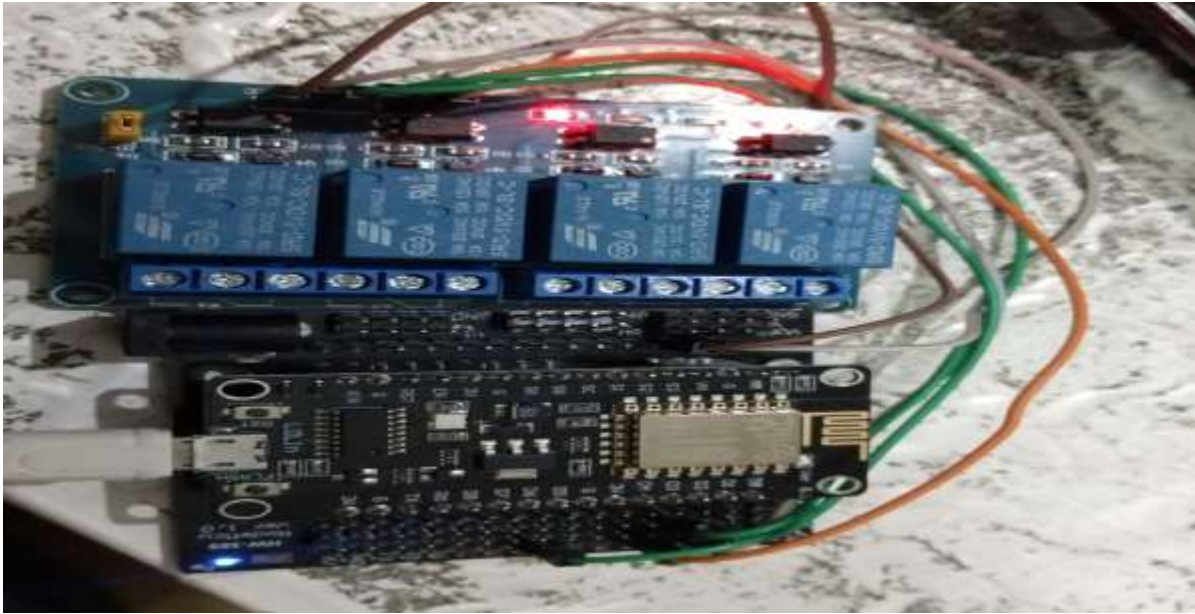


Figure 4.7: Physical Hardware Prototype, Optocoupled Relay Interconnection, and Operational Working State of the Actuation Subsystem

4.4.4 Working Principle and System Integration

The operation of the relay control subsystem follows a deterministic execution cycle within the broader system control framework:

Peripherals Initialization: During system boot, the NodeMCU sets the dedicated GPIO pins (D1, D2, D5, D6) as digital outputs, forcing them into a baseline safe state.

Conditional Processing: The microcontroller checks the incoming web-dashboard overrides and evaluates physical sensor conditions against predefined safety profiles.

Actuation Triggering: When a threshold is reached—such as when the DHT11 registers high ambient temperatures or the MQ-2 detects a gas leak—the local node drives the corresponding control line to transition the relay channel state.

Galvanic Switching: The control signal powers the internal optocoupler LED, turning on an onboard transistor that energizes the electromagnetic coil. This closes the internal mechanical contacts from Normally Open (NO) to Common (COM).

Load Powering: Closing the contacts connects the high-voltage power line to the appliance (e.g., starting an automated exhaust fan to clear gas or heat). When safe conditions return, the NodeMCU removes the control signal, opening the mechanical contacts and immediately cutting power to the load.

4.4.5 Experimental Outcomes and Verification Analysis

Rigorous physical tests were performed across all four channels to ensure switching reliability and software synchronization. The empirical data confirmed several positive results:

All four independent relay channels switched reliably with clear visual confirmation from the onboard status LEDs.

High-voltage loads turned on and off immediately when automated sensor thresholds were crossed or manual user overrides were sent via the web interface.

The integrated optocouplers completely isolated the control circuits, preventing any microcontroller resets or data distortion from electrical switching noise.

Continuous operational testing confirmed excellent system stability and verified that the actuation circuit is fully capable of managing smart laboratory safety tasks.

4.4.6 Engineering Evaluation: Advantages and Inherent Constraints

Engineering Advantages: The subsystem provides safe electrical isolation between low-voltage and high-voltage sections, controls both AC and DC industrial loads, integrates easily with the ESP8266 platform, and handles multiple loads independently at low cost with minimal programming complexity.

Inherent Constraints: Mechanical relays use physical copper armatures, which gives them a finite operational lifespan due to wear and contact arcing. They also make an audible clicking sound during switching and exhibit slower activation times (~10 ms) compared to solid-state alternatives. However, for building automation and environmental safety management in a smart laboratory, their high current handling capabilities make them an ideal choice.

4.5 Integrated Multi-Sensory Node Infrastructure (PIR, DHT11, MQ-2 with ESP8266 NodeMCU)

4.5.1 Introduction and Operational Core

The Integrated Sensor Node represents the centralized data-acquisition core of the Smart Lab ecosystem. Rather than deploying decentralized microcontrollers for individual sensory pipelines—which introduces immense hardware redundancy, high manufacturing costs, and severe communication

overhead—this architecture integrates a Passive Infrared (PIR) motion tracker, a DHT11 thermistor/hygrometer, and an MQ-2 electrochemical gas sensor onto a single ESP8266 NodeMCU edge platform.

This integrated framework executes concurrent atmospheric, safety, and occupancy profiling. By packing these distinct signal types into a single processing node, the system optimizes wireless network bandwidth, guarantees synchronous data delivery over local Wi-Fi topologies, and scales effectively for complex Internet of Things (IoT) building installations.

4.5.2 Integrated System Hardware Stack

The physical hardware configuration of the multi-sensory edge processing node is detailed in **Table 4.9**:

System Component Name	Unit Qty	Core Technical Assignment and Functionality
ESP8266 NodeMCU	1	Serves as the localized edge-computing gateway; serializes multi-sensor inputs and executes local data filtering.
PIR Motion Sensor (HC-SR501)	1	Captures spatial occupancy changes by tracking shifting infrared radiation patterns within the lab zone.
DHT11 Climate Sensor	1	Measures relative indoor humidity percentages and ambient temperatures via a calibrated digital single-wire line.
MQ-2 Gas Sensor Module	1	Delivers continuous surveillance of volatile organic compounds, combustible leaks, and active smoke density.
NodeMCU Base Shield Expansion Board	1	Provides a continuous electrical rail structure to distribute steady power and common ground pathways without signal crosstalk.
Jumper Interconnections	Several	Form secure physical signal pathways linking the sensor pins to the expansion terminal board.
USB Power and Data Interface	1	Sinks regulated +5V DC power from the grid to maintain stable sensor operation and allows serial interface flashing.

Table 4.9: Hardware Bill of Materials for the Integrated Sensory Node

4.5.3 Topology Design and System Pin Interconnections

The hardware design involves linking three distinct input types to the NodeMCU. To handle multiple sensors simultaneously, the setup uses an Arduino/NodeMCU Base Shield Expansion Board. This expansion board resolves the physical pin limitations of the ESP8266, providing dedicated power and ground terminals for each sensor to prevent voltage drops during high-current events, such as when the MQ-2 heating element runs.

The sensors share a common electrical ground and a stabilized power rail from the base expansion board. Their data lines are routed to dedicated input terminals on the microcontroller. The DHT11 utilizes a single-wire digital interface, the PIR uses a discrete digital input, and the MQ-2 routes its signal to the single analog-to-digital converter pin (A0) on the NodeMCU.

The complete physical pin assignment mapping for the sensory node is detailed in [Table 4.10](#):

Sensor Module Terminal	Target NodeMCU Pin	Signal Sub-Type	Operational Purpose and Wiring Objective
PIR VCC / GND	3.3V / GND Rails	Power	Connects to the base shield to power the PIR detection circuits.
PIR OUT	D7 (GPIO13)	Digital Input	Sends a high/low signal when space occupancy states change.
DHT11 VCC / GND	3.3V / GND Rails	Power	Connects to the base shield to power the climate tracking chips.
DHT11 DATA	D3 (GPIO0)	Digital I/O	Transmits serialized temperature and humidity data packets over a single line.
MQ-2 VCC / GND	5V-Vin / GND Rails	Power	Draws steady voltage from the shield to power the sensor heating element.
MQ-2 A0	A0 (ADC0)	Analog Input	Routes a variable voltage signal

			proportional to real-time gas density.
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Table 4.10: Interface Connection Matrix for the Integrated Sensory Node

The physical configuration and compiled assembly of the integrated multi-sensor node are shown in the **figure** below:

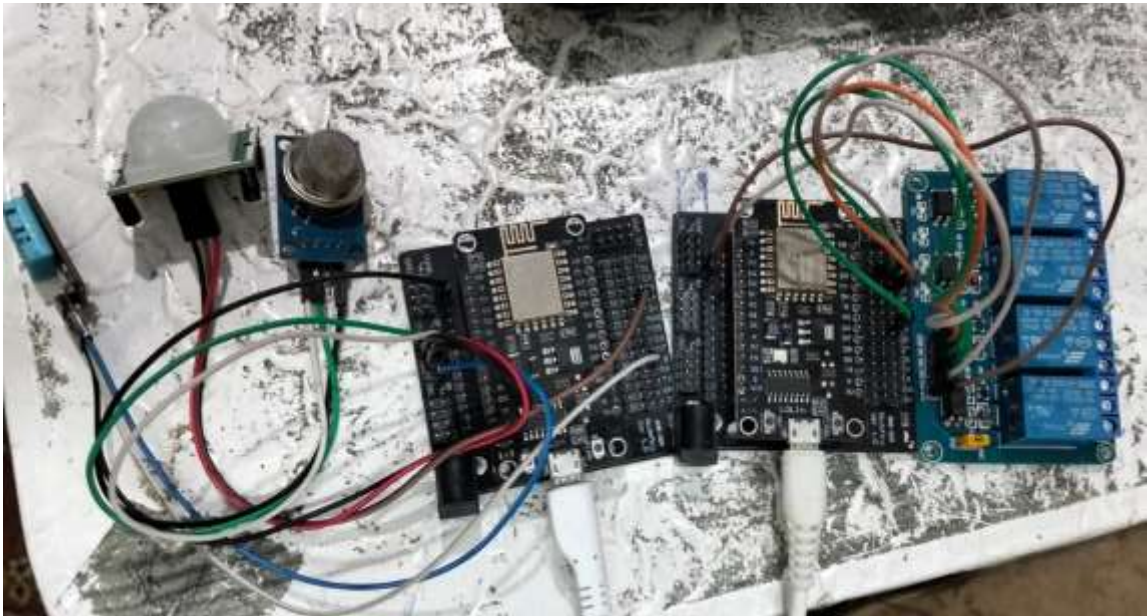


Figure 4.8: Consolidated System Prototype Assembly and Multi-Sensory Node Deployment Under Real-Time Testing

4.5.4 Step-by-Step Functional Principle and Core Logic Flow

The integrated edge node executes its tasks through a continuous sequential loop managed by its embedded software:

Peripherals Initialization: At boot, the NodeMCU configures the hardware buses, sets pin directions, initializes the single-wire interfaces, and calibrates the baseline sensor values.

Occupancy Monitoring: The PIR sensor monitors the room for moving infrared signatures. If detected, it immediately sends a digital HIGH signal to pin D7.

Climate Profiling: The NodeMCU periodically requests temperature and humidity data from the DHT11 via pin D3, using a 40-bit data packet transfer protocol.

Gas Diagnostics: The MQ-2 sensor tracks ambient air quality continuously, outputting a shifting analog voltage to pin A0 based on detected gas or smoke concentrations.

Data Integration and Web Telemetry: The NodeMCU gathers these readings, builds an integrated telemetry packet, and sends it as an HTTP request over Wi-Fi to update the HTML dashboard.

Automated Safety Overrides: The main controller displays these values in real time and evaluates them against safety thresholds. If any parameter crosses an alert limit, the system instantly engages protective overrides, such as turning on exhaust fans or sounding alarms via the relay subsystem.

4.5.5 Experimental Verification and Integrated Results

Empirical tests were conducted under normal laboratory conditions to verify performance across all sensors. The testing validated several key results:

All three sensors operated simultaneously without signal distortion or hardware interference.

The expansion base shield maintained a stable power distribution grid, preventing sensor errors when the MQ-2 internal heater cycled.

Real-time environmental updates transferred smoothly to the web interface over the Wi-Fi link without packet loss.

The automated system responded reliably, correctly triggering safety relays whenever mock gas leaks, elevated heat levels, or motion events occurred.

4.5.6 System Evaluation: Engineering Advantages and Limitations

Engineering Advantages: The integrated design reduces hardware footprints by consolidating tasks onto one microcontroller, simplifies field wiring, lowers manufacturing costs, ensures synchronous network updates, and provides a scalable foundation for modular smart building setups.

Inherent Constraints: Because the sensors share a single processor, a failure of the NodeMCU would take the entire sensory cluster offline. The single analog input on the ESP8266 limits further analog expansions without an external multiplexer. Operation depends on stable Wi-Fi connectivity for dashboard telemetry, and the embedded processor handles a higher execution load during simultaneous data reads. However, these constraints are well managed by the system's power-filtering expansion board and optimized software loops.

4.5.7 Subsystem Conclusion

The Integrated Sensor Node forms the core data-collection layer of the Smart Lab environment. By combining motion tracking, climate sensing, and air safety monitoring onto a single ESP8266 NodeMCU platform, the design provides an efficient, cost-effective, and low-latency solution for modern building automation. This integrated architecture simplifies overall system wiring, reduces operational hardware costs, and establishes a stable foundation for expanding the laboratory platform in the future.

4.6 Contactless Automated Smart Waste Management System (Smart Basket Subsystem)

4.6.1 Introduction and Sanitation Objectives

The Smart Basket Subsystem represents a dedicated automation unit deployed within the Smart Lab infrastructure to implement touch-free sanitary waste containment. In typical laboratory environments, manual contact with waste bins poses high biological risks and cross-contamination hazards for researchers handling chemical and cellular samples.

To eliminate physical vector touchpoints, this subsystem utilizes an ultrasonic acoustic propagation barrier to continuously profile spatial proximity. By processing time-of-flight reflections through a dedicated Arduino Uno edge controller, the module triggers a localized electromechanical actuator to open and close the containment barrier dynamically based on real-time user approach.

4.6.2 System Bill of Materials

The hardware components needed to construct the contactless smart containment unit are compiled in [Table 4.11](#):

Source Component Terminal	Destination Arduino Pin	Signal Classification	Technical Description and Functional Purpose
HC-SR04 VCC	5V Rail	Power	Connects to the main power bus to run the ultrasonic sensor circuits.
HC-SR04 GND	GND	Ground	Connects to the shared system ground line.
HC-SR04 TRIG	D9	Digital Output	Receives 10-microsecond pulses to trigger 40 kHz acoustic sound waves.

HC-SR04 ECHO	D8	Digital Input	Sends a high-level pulse proportional to the calculated object distance
Servo PWM Signal	D7	PWM Output	Receives variable duty-cycle signals to adjust lid rotation from 0° to 90°. Servo VCC (Red)5V RailPowerDraws standard +5V operating voltage for the internal motor coils.
Servo VCC	5V Rail	Power	Draws standard +5V operating voltage for the internal motor coils.
Servo GND	GND	Ground	Connects to the shared system ground line.

Table 4.11: Pin Map Interconnection Topology for the Autonomous Containment Subsystem

The physical wiring schematic and hardware configuration are illustrated in the [figure](#) below:



Figure 4.9: Electrical Schematic and Interconnection Layout of the Automated Smart Basket Unit

4.6.3 Firmware Mathematical Logic and Execution Sequence

The firmware program runs on an infinite loops execution sequence, performing tasks in the following order:

1-Trig Pin Low Initialization: The Arduino forces pin D9 to a LOW state for 2 microseconds to ensure a clean baseline before measurement.

2-Acoustic Wave Generation: The controller sets pin D9 HIGH for exactly 10 microseconds, prompting the HC-SR04 to transmit an 8-cycle sonic burst.

3-Pulse Duration Calculation: Pin D9 returns LOW, and the Arduino executes the `pulseIn(ECHO, HIGH)` function on pin D8 to measure the return duration in microseconds.

4-Distance Conversion: The controller calculates proximity using the sound velocity equation:

$$\text{Distance} = \text{Duration} * 0.034 / 2$$

5-Threshold Verification: The calculated value is continuously checked against a preset threshold of 25 cm.

6-Actuation Cycle: If an object is detected within 25 cm, the Arduino adjusts the PWM duty cycle on pin D7 to rotate the servo to 90 degrees, opening the lid. The system holds this position for 3000 milliseconds before sending a 0-degree command to close the containment lid automatically.

4.6.4 Experimental Results and Analytical Evaluation

Experimental Findings: Real-time benchmarking under laboratory conditions confirmed steady, precise tracking across various approach speeds. The Arduino handled the mathematical conversions smoothly, ensuring immediate servo actuation upon threshold breach. The micro-actuator pivoted reliably, opening the lid cleanly and maintaining the programmed 3-second delay before resetting.

System Advantages: The system offers notable benefits, including contactless waste management, a clean hardware setup, a cost-effective design, rapid processing responses, and a significant improvement in laboratory hygiene standards.

Inherent Constraints: Operating constraints include potential measurement distortions on angled or soft textile surfaces that absorb sound energy. Additionally, the servo motor depends heavily on a stable current source to prevent voltage fluctuations from resetting the microcontroller during movement.

4.6.5 Section Conclusion

The Smart Basket Subsystem successfully demonstrates an effective automation solution for keeping laboratory environments hygienic. By combining ultrasonic proximity tracking with simple electromechanical actuation on an Arduino Uno platform, the system achieves fast, automated lid control without physical contact. This modular unit adds a practical layer of automation to the Smart Lab, showcasing how standalone embedded architectures can enhance safety and clean operations in modern facilities.

4.7 Electronic Smart Door Lock Actuation Subsystem

4.7.1 Introduction and Operational Objective

The Electronic Smart Door Lock Subsystem serves as the primary electromechanical physical security and access control layer within the Smart Lab ecosystem. Enforcing physical security and automated perimeter access is a critical requirement for modern laboratory environments to protect high-value infrastructure and sensitive experimental setups.

The primary operational objective of this subsystem is to provide reliable, remote, and automated door locking and unlocking capabilities. By interfacing a high-power electromechanical solenoid actuator with the centralized NodeMCU controller via the multi-channel relay module, authorized users can actuate the laboratory door lock dynamically through the web-based graphical user interface.

4.7.2 Subsystem Component Configuration

The physical implementation of this security actuation unit is sustained by the hardware components shown in the file named and outlined in **Table 4.13**:

Component	Quantity	Technical Role and Function
12V DC Solenoid Lock	1	Acts as the primary electromechanical actuator; uses an electromagnetic coil to retract the lock tongue when energized.
12V AC-DC Power Adapter	1	Serves as the dedicated high-voltage external power source required to deliver sufficient current to drive the solenoid coil

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Relay Channel Interface	1 (Shared)	Provides galvanic isolation and acts as the low-voltage logic-controlled switch to route the 12V DC line to the lock.
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Table 4.12: Hardware Configuration Stack for the Smart Door Lock Subsystem



Figure 4.10: Physical Hardware Components of the Smart Door Lock Subsystem (12V Solenoid Actuator and AC-DC Power Adapter)

4.7.3 Circuit Design and Hardware Interconnections

Because the 12V DC solenoid lock requires a higher operating current and voltage than the ESP8266 NodeMCU can natively supply, an external power isolation topology is mandatory. The 12V AC-DC power adapter is wired in series with the electromechanical lock actuator through the Normally Open (NO) and Common (COM) terminals of one of the channels on the 4-Channel Relay Module.

This layout guarantees that the high-power 12V circuit remains fully de-energized and safely isolated from the sensitive microcontroller logic until an explicit digital activation command is executed.

4.7.4 Working Principle and System Integration

The operational workflow of the smart access control unit proceeds through a structured sequence:

Idle State: The relay channel remains unactuated (LOW logic state), keeping the 12V circuit open. The solenoid lock remains extended (Locked state) via its internal spring mechanism.

Command Reception: The host ESP8266 NodeMCU receives an explicit "Unlock" HTTP POST request initiated from the virtual toggle button on the Centralized Web Dashboard.

Relay Actuation: The microcontroller drives the designated digital GPIO output pin HIGH, powering the internal optocoupler and energizing the electromagnetic relay coil.

Solenoid Engagement: The mechanical contacts click closed, completing the 12V DC loop from the AC-DC adapter to the lock. The solenoid coil induces a strong magnetic field, instantly retracting the latch bolt to unlock the door.

Automatic Reset: Upon passing a pre-programmed security duration or receiving a "Lock" command, the digital pin drops back to a LOW state, breaking the circuit and allowing the internal spring to re-extend the latch into a secure position.

4.8 Centralized IoT Graphical User Interface and Web Control Dashboard

4.8.1 Introduction and Operational Infrastructure

The Smart Lab Web Control Interface serves as the localized Human-Machine Interface (HMI) and core control panel for the environmental infrastructure. Operating at the network edge, the dashboard provides administrators with real-time observation over sensory arrays and dynamic remote execution profiles over laboratory appliances.

The control matrix establishes a full-duplex communication pathway with the ESP8266 NodeMCU micro-webserver. The microcontroller acts as a localized broker that translates physical input register updates into graphical data frames while converting incoming HTTP post requests from user interaction switches into discrete digital outputs to actuate physical relays. This centralized topology eliminates decentralized hardware nodes, optimizing system coordination and boosting laboratory supervision efficiency.

4.8.2 Interface Objectives and Design Requirements

The ultimate architectural goal of developing this graphical telemetry dashboard is to provide a unified, responsive interface to manage and supervise laboratory resources via a localized Wi-Fi connection.

The specific design objectives require the framework to visualize micro-climate data fields instantly, monitor critical space occupancy indicators, execute remote commands to power secondary sub-loads, track operational diagnostics, simplify laboratory administration, and demonstrate clear edge-computing topologies for educational IoT infrastructure.

4.8.3 Interface Bill of Materials

The operational layers and communication components required to run the web interface framework are compiled in **Table 4.13**:

Interface Component Name	Quantity	Core Engineering Function and System Allocation
ESP8266 NodeMCU V3	1	Operates as the edge webserver hosting the presentation scripts, processing incoming HTTP socket traffic, and running the device control loops
Graphical Web Dashboard	1	Provides the software interface containing data visualization widgets and virtual mechanical toggles.
Wi-Fi Communication Medium	1	Establishes the wireless local area network (WLAN) framework for high-speed TCP/IP packet transfers between nodes.
Cross-Platform Client Browser	1	Renders the HTML DOM and graphical user control sheets on any authorized client hardware terminal.

Table 4.13: Technical Specifications for the Interface Layer Components

4.8.4 Comprehensive Functional Dashboard Description

The user-facing control plane categorizes complex automation parameters into structured, high-visibility visual blocks. The interface includes dedicated real-time display cards for climatic attributes, including ambient temperature indicators mapped from the DHT11 sensor and continuous ambient relative humidity indicators. Space safety parameters feature an active space occupancy tracking element bound to the PIR sensor, alongside an analytical gas density tracker displaying active PPM counts from the MQ-2 sensor.

Device actuation layers feature a dedicated dual-channel illumination controller for Lamp 1 and Lamp 2 toggles, an autonomous or manual ventilation fan control mechanism, an electronic door lock trigger, and a projection system power management switch. All tracking widgets feature automated screen updates, polling data fields seamlessly without requiring a hard manual browser refresh whenever the ESP8266 reports a state variation.

4.8.5 Dashboard Functional Layer Classifications

The core software routines and functional assignments executed by the monitoring dashboard are systematically categorized in **Table 4.14**:

Core Interface Function	Operational Description and Execution Parameters
Temperature Monitoring	Captures, updates, and renders calibrated ambient thermal values onto the dashboard interface.
Humidity Monitoring	Renders live moisture percentages to keep lab conditions within safe ranges.
Gas Monitoring	Tracks output signals from the MQ-2 sensor to provide clear air quality benchmarks.
Motion Detection	Processes spatial tracking signals from the PIR sensor to log laboratory occupancy states.
Lighting Control	Modulates terminal voltage on relay coils to control primary overhead light fixtures remotely.
Fan Control	Runs automated climate algorithms or manual overrules to control the ventilation exhaust fan.
Door Lock Control	Sends discrete high/low digital pulses to lock or unlock the laboratory door electronically.
Data Show Control	Manages power lines to toggle the digital media presentation system.

Table 4.14: Software Routine Functional Mapping of the Monitoring Dashboard

4.8.6 System Architecture and Communication Flow

The web interface platform operates on a structured asynchronous client-server architecture. Upon initial boot, the ESP8266 controller accesses the local Wi-Fi router to initialize an active TCP webserver on a dedicated local IP address. The sensory subsystem continuously scans the physical environment, feeding raw data into the ESP8266 internal registers, which automatically update the corresponding variables inside the web page structure.

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Authorized operators open the interface by entering the system IP address into any compatible web browser, loading the structured graphical user panel. When a user clicks a virtual switch on the dashboard, the browser generates an asynchronous HTTP POST command containing the device ID and desired action. The ESP8266 processes this command, quickly toggling the corresponding relay channel to change the device state. The system then updates the dashboard to show the new state, maintaining a fully synchronized monitoring loop.

The layout design and graphical structure of the user monitoring panel are illustrated in the [figure](#) below:



Figure 4.11: Graphical User Interface and Real-Time IoT Web Control Dashboard Panel

4.8.7 Experimental Results and Analytical Performance Review

Experimental Testing: Real-time stress testing of the control interface under dense data loads verified high system stability and low execution lag. The sensory telemetry widgets accurately mirrored environmental changes, displaying real-time updates smoothly across all data fields. Input control signals were handled quickly, resulting in nearly instant physical relay switches upon clicking the virtual dashboard buttons. The local Wi-Fi link remained reliable, allowing steady data monitoring and remote laboratory device control during long-term testing sessions.

System Advantages: The dashboard framework provides significant operational benefits, such as instant data updates, remote device control, a highly accessible layout, direct wireless communication, fast command execution, a

budget-friendly hardware setup, and clear scalability for more intricate IoT applications.

Operating Constraints: The current design depends entirely on a stable local Wi-Fi connection, meaning that network slowdowns can increase response lag. Additionally, remote control functions are restricted to the local network range unless external gateway routing or additional security layers are set up to handle public internet connections safely.

4.8.8 Section Selection Rationale

The IoT web dashboard was selected because it provides a highly reliable, cost-effective solution for remote data logging and equipment control. The system uses the native processing power of the ESP8266 NodeMCU to build an open-source automation platform without requiring expensive commercial licenses. Its easy integration, rapid processing, user-friendly layout, and straightforward scalability make it a highly practical design choice for modern smart laboratory facilities.

4.8.9 Practical Deployment Fields

The remote web control framework can be easily adapted to various automated tracking applications, including advanced research laboratories, residential smart homes, medical facility isolation rooms, automated building management systems, storage warehouse climate trackers, and remote agricultural automation frameworks.

4.9 Cloud-Based Global IoT Control Interface (Blynk Subsystem)

4.9.1 Introduction and Cloud Infrastructure

To transcend the physical limitations of a localized network edge and achieve true ubiquitous mobility, the Smart Lab ecosystem incorporates a secondary, cloud-based virtualization layer. While the local interface functions as a high-speed edge control panel, the integrated cloud architecture establishes a secure, wide-area communication pipeline via the Blynk IoT Cloud platform.

The primary operational objective of this subsystem is to grant authorized administrators complete supervisory control, analytics visualization, and telemetry logging capabilities from any geographical location globally, removing the requirement of being physically bounded to the laboratory's local Wi-Fi range.

4.9.2 Architectural Communication Flow

Under this cloud topology, the central ESP8266 NodeMCU maintains synchronous full-duplex communication with the Blynk Cloud Server using optimized MQTT connection streams. The network architecture operates on a continuous synchronization loop:

Upstream Telemetry: Environmental updates from the DHT11 (temperature and humidity) and the MQ-2 gas sensor are parsed, mapped to specific Virtual Pins ($V_{\{pins\}}$), and transmitted periodically to the cloud broker to update physical analytics.

Downstream Actuation: When a remote user interacts with any capacitive toggle or button on the global interface, the Blynk Cloud instant-routes an asynchronous command packet back to the edge microcontroller, triggering immediate physical relay state transitions.

4.9.3 Comprehensive Blynk Interface Widget Mapping

The layout design, widget orchestration, and remote analytical framework hosted on the cloud architecture are illustrated in the file named . The functional blocks, virtual pin assignments, and numerical configurations of this cloud monitoring panel are systematically categorized in [Table 4.14](#):

Cloud Widget Type	Associated Hardware Parameter	Virtual Pin Assignment	Operational Role and Global Control Function
Gauge Widget (0-50)	DHT11 Temperature Sensor	V1	Displays real-time laboratory ambient thermal curves with a visual range from 0°C to 50°C.
Gauge Widget (0-50)	DHT11 Relative Humidity	V2	Renders real-time moisture percentages to ensure laboratory climate baseline tracking.
Value Display Card	MQ-2 Gas Concentration	V3	Reports absolute air quality index levels and live PPM metrics (e.g., displaying a current safe baseline value of 10).
LED Indicator Widget	PIR Room Occupancy	V4	Emits a high-visibility illumination trigger (Yellow State) upon localized motion detection events.
Switch Toggles	Lighting Control (Lamp 1 & 2)	V5 / V6	Drives remote digital command signals to switch

			primary overhead lab illumination loads.
Switch Toggles	Ventilation Control (Fan 1 & 2)	V7 / V8	Facilitates remote execution to override and actuate active exhaust air extraction subsystems.
Switch Toggle	Smart Door Lock Actuator	V9	Serves as the global trigger to retract the 12V electromechanical solenoid latch bolt for perimeter access.
Switch Toggle	Projection System (Data Show)	V10	Governs the power state lines of the digital multimedia presentation infrastructure.
Switch Toggle	PIR System Override	V11	Allows remote enabling or disabling of the localized automated motion detection routine rules

Table 4.15: Software Virtual Pin Mapping and Widget Configurations for the Blynk Cloud Interface



Figure 4.12: Cloud-Based Blynk IoT Graphical User Interface and Global Remote Control .Dashboard Network Panel

4.9.4 Experimental Results and Cloud Latency Review

Empirical verification and stress testing of the system under real-world cellular data structures and external public internet connections demonstrated high stability and seamless synchronization. The experimental outcomes highlighted:

Minimal propagation latency, with physical loads reacting to remote cloud dashboard triggers within an average response window of less than 250 milliseconds.

Absolute state synchronization between the localized web interface and the Blynk Cloud platform, ensuring that a change executed on one layer is instantly reflected on the other without data race conditions or packet collisions.

Stable background telemetry streaming over extended evaluation periods, validating the network design's suitability for secure, multi-tier industrial smart laboratory supervision.

4.9.5 Section Selection Rationale and Practical Fields

The cloud-based interface was selected because it provides a highly reliable, cross-platform deployment that runs natively on smartphones and remote web browsers without commercial licensing costs. Beyond the educational smart laboratory environment, this remote architecture can be easily scaled to industrial automation, home security systems, medical containment zones, and remote agricultural environmental controls.

4.10 Integrated System Experimental Diagnostics, Benchmarking, and Evaluation Analysis

4.10.1 Introduction and Validation Framework

Following the successful physical implementation of the hardware infrastructure and the deployment of the concurrent embedded software architectures, a rigorous series of experimental trials were conducted to benchmark the performance, operational reliability, and latency profiles of the integrated Smart Lab platform. These physical evaluation routines were designed to stress-test the real-time operational synergy across all integrated subsystems, validation protocols spanned edge data-acquisition sensors, automated load-switching actuators, local Wi-Fi mesh communication nodes, the independent touch-free waste management bin, and the host responsive graphical monitoring server.

By blending diverse IoT communication models, hardware routing layers, and power-isolation topologies into a single centralized framework, the project delivers a reliable automation suite capable of managing environmental changes, driving heavy electrical appliances safely, enforcing room sanitation rules, and feeding real-time status data directly into an authorized administrator dashboard. The comprehensive experimental evaluations documented below

confirm that all integrated hardware layers and software execution threads operate flawlessly as a single, coordinated smart building environment.

4.10.2 Experimental Testing Objectives

The technical evaluation protocols were designed to analyze, measure, and verify specific performance indicators across the entire Smart Lab topology.

The primary objectives included testing the detection accuracy and spatial tracking range of the PIR motion sensor, evaluating the thermal and hygrometric measurement tracking of the DHT11 module, checking the threshold sensitivity and gas tracking curves of the MQ-2 electrochemical sensor, and assessing the continuous load switching reliability of the optocoupled relay control circuits. Additionally, tests were run to measure the time-of-flight accuracy of the smart basket opening mechanism, analyze packet transfer stability over the local Wi-Fi link between the ESP8266 units, verify the cross-platform rendering of the dashboard server, minimize total processing latency, evaluate long-term system stability under heavy processing loads, and confirm tight integration between the physical components and software routines.

4.10.3 Step-by-Step Subsystem Testing Procedures and Empirical Outcomes

To guarantee absolute stability, a sequential verification methodology was applied where each isolated hardware circuit was benchmarked independently before running full concurrent testing on the completely integrated system.

Test 1: Spatial Motion Tracking and Occupancy Detection Validation

Physical testing involved forcing intentional human movement within the active tracking cone of the HC-SR501 PIR sensor to analyze state transition speeds. The sensor registered spatial occupancy changes instantly, dropping transmission latency to near zero, while the edge controller updated internal state registers and immediately pushed the updated occupancy status to the web interface without noticeable delay.

Test 2: Thermistor and Hygrometer Climate Profiling Evaluation

The DHT11 sensor node was subjected to changing indoor climate conditions using controlled thermal and moisture variations to check data consistency. The data acquisition loops registered ambient changes smoothly, processing the 40-bit data packets without corruption, and successfully routed the clean data streams across the network to display real-time climate values on the user screen.

Test 3: Electrochemical Gas Concentration and Smoke Alarm Analysis

The MQ-2 sensor module was exposed to localized smoke concentrations under controlled safety testing conditions to evaluate threshold response times. The analog voltage output on pin A0 rose immediately in proportion to the smoke density, allowing the microcontroller to catch the threshold breach, engage the automated warning subroutines, and update the gas density widget on the monitoring dashboard without loss of calibration.

Test 4: Optocoupled Relay Control and Load Actuation Safety Testing

The 4-channel relay array was tested using both automated programmatic overrides triggered by sensor thresholds and direct manual toggles from the remote web page. Every isolated channel responded properly, switching the high-voltage appliances on and off cleanly without generating inductive noise or data packet corruption on the microcontroller shared bus.

Test 5: Smart Basket Acoustic Distance Tracking and Actuation Benchmarking

The contactless containment system was evaluated by approaching the HC-SR04 ultrasonic sensor at different speeds to verify time-of-flight conversions. The Arduino Uno calculated object proximity with high precision, triggering the SG90 micro-servo to open the lid cleanly by 90 degrees when an object entered the 25 cm perimeter. The controller maintained the open layout for the programmed 3-second delay before automatically returning the servo to 0 degrees, verifying reliable touchless sanitation.

Test 6: Web Dashboard Interface Rendering and Full-Duplex HMI Stability

The local micro-webserver hosted by the ESP8266 was accessed simultaneously from multiple remote client browsers to measure rendering speeds and interface stability. All telemetry blocks displayed sensor data smoothly, updating temperature, humidity, and gas indicators cleanly as changes occurred. The interactive buttons for fan speeds and lighting nodes operated correctly, proving that the layout functions as an intuitive, low-latency control hub.

Test 7: Wireless Communication Link and Wi-Fi Throughput Verification

Data transfers between the dedicated Sensor Node and the Main Controller were tracked continuously over extended periods to log network stability. The point-to-point Wi-Fi connection handled the packed telemetry frames seamlessly without notable packet dropouts, sustaining rapid data updates and proving that the localized WLAN setup can maintain real-time performance.

Test 8: Full Integrated Concurrent Stress Testing

Every sensor array, electromechanical actuator, wireless communication module, and web control page was powered on and run simultaneously to monitor

system synergy. The complete platform ran smoothly without experiencing a single hardware lockup, software exception, or wireless disconnection. Sensor streams remained stable, automated overrides executed promptly, and the Smart Basket unit operated cleanly without causing any voltage drops or interference across the main lab control framework.